Rice Farming Handbook

By NOMAD

Rice farming is the backbone of Philippine agriculture, as rice is the primary staple food for over 90% of the population [21]. The country is the world's 8th-largest producer, yet it remains one of the largest importers to meet its high domestic demand.

Core Industry Statistics (as of 2020–2025)

- **Total Farmers:** Approximately 2.4 to 2.5 million farmers.
- **Average Farm Size:** Continues to decline, currently estimated at less than 1.4 hectares.
- **Average Farmer Age:** 56 years old, with 82% being male.
- **Production Volume:** Roughly 19.1 to 20 million metric tons of *palay* (unmilled rice) annually.
- **Key Regions:** Major production centers include **Cagayan Valley**, **Central Luzon** (the "Rice Granary"), **Ilocos**, **Western Visayas**, and **Mindanao**.

Farming Methods & Ecosystems

Rice in the Philippines is grown year-round across three primary ecosystems [4, 6, 18, 35]:

1. **Irrigated Lowland (70%):** Areas with a consistent water supply through canals, often allowing for 2–3 crops per year [6, 18].
2. **Rainfed Lowland (30%):** Fields that depend entirely on rainfall, common in areas like Mindanao
3. **Upland/Terrace:** Ancient practices in mountainous regions like the **Banaue Rice Terraces**, where traditional and heirloom varieties are often preserved [3, 37, 40].
4.

The 10-Step Production Cycle

Rice production typically follows these stages:

1. **Seed Selection:** Choosing varieties (e.g., *NSIC Rc 216*, *Rc 160*) based on climate resilience.

2. **Land Preparation:** Plowing and harrowing (often using a *carabao* or hand tractor) and leveling the soil.
3. **Crop Establishment:** Can be done via **transplanting** (moving nursery seedlings to the main field) or **direct seeding**.
4. **Water Management:** Flooding paddies 5–10 cm deep or using **Alternate Wetting and Drying (AWD)** to save water.
5. **Fertilizer Application:** Applying organic and chemical nutrients at critical growth stages.
6. **Pest & Disease Management:** Monitoring for "Rice Tungro" or "Brown Planthoppers".
7. **Harvesting:** Cutting mature stalks when grains reach 20–25% moisture.
8. **Threshing:** Separating the grains from the straw.
9. **Drying:** Reducing moisture to 14% to prevent spoilage.
10. **Milling:** Removing the husk and bran to produce white rice (*bigas*).

Key Challenges & Innovations

- **Climate Change:** Frequent typhoons, droughts (El Niño), and floods cause significant crop losses annually.
- **High Costs:** High prices for fertilizers, labor, and fuel often result in low profit margins for smallholders.
- **Modernization:** Organizations like the International Rice Research Institute (IRRI) and Philippine Rice Research Institute (PhilRice) are promoting hybrid seeds and Precision Agriculture using drones.
- **Policy:** The **Rice Tariffication Law** aims to modernize the industry through the Rice Competitiveness Enhancement Fund (RCEF), though its impact on small-scale farmers remains a subject of debate.

Financial Profitability (Per Hectare, 2024 Estimates)

According to the **Philippine Statistics Authority (PSA)**, the average financial performance of a one-hectare rice farm per cropping season is as follows:

- **Gross Returns:** Approximately **₱95,906**, a 16% increase from the previous year due to higher farmgate prices.
- **Total Production Cost:** Roughly **₱59,695** (approx. **₱14.52 per kilogram**).
- **Labor:** The largest expense, averaging **₱29,235** per hectare.
- **Seeds:** Average cost of **₱4,316**.
- **Net Returns:** Average of **₱36,211** per hectare.
- **Top Performing Region:** Central Luzon recorded the highest net returns at **₱58,932** per hectare.
- **Farmgate Price:** Averaged **₱23.48 per kilogram** of *palay* (unmilled rice) in 2024.

Government Subsidies and Assistance

The Philippine government provides several aid programs primarily through the Department of Agriculture (DA) and the Rice Competitiveness Enhancement Fund (RCEF).

- **Cash Assistance (RFFA):** Smallholder farmers tilling **2 hectares or less** are eligible for unconditional cash aid, typically **₱5,000 to ₱7,000** per year.
- **Input Subsidies:**
- **Certified Seeds:** Free distribution of high-quality inbred seeds through the RCEF Seed Program.
- **Fertilizer Vouchers:** Distributed to help offset the rising costs of chemical fertilizers.
- **Mechanization:** Farmer cooperatives can receive free farm machinery (tractors, harvesters, dryers) from the PHilMech RCEF Mechanization Program to reduce labor costs.
- **Credit Programs:** The Agri-Puhunan at Pantawid (APP) Program provides low-interest loans for seeds and fertilizers, often paired with a guaranteed buying price for the harvest.
- **Consumer-Facing Subsidies:** Programs like **KADIWA** centers sell rice at subsidized rates (e.g., **₱20-₱29 per kilo**) to vulnerable sectors.

Choosing between hybrid and inbred rice involves balancing high potential yields against significantly higher upfront costs

To access most government support, farmers must be registered in the **Registry System for Basic Sectors in Agriculture (RSBSA)**.

Hybrid vs. Inbred Rice: A Comparison

Feature	Hybrid Rice	Inbred Rice (Certified/High Quality)
Yield Potential	**7–15 metric tons** per hectare.	**3.6–6 metric tons** per hectare.
Seed Cost	**₱5,000–₱7,320** per hectare (~₱250–₱400/kg).	**₱1,150–₱2,400** per hectare (~₱30–₱40/kg).
Seed Quantity	Only **15–20 kg** needed per hectare.	**40–80 kg** needed per hectare.
Re-planting	**Cannot be re-planted**; new seeds must be bought every season to maintain yield.	**Can be re-planted** for 2–3 seasons if handled carefully.
Management	Requires **intensive management**, more fertilizer, and precise water control.	**Stable and predictable**; generally requires fewer chemical inputs.
Net Profit	Can exceed **₱70,000–₱150,000** per hectare in optimal conditions.	Typically lower per hectare, but has **higher benefit-cost ratios** in low-input settings.

Eligibility for Subsidies & Assistance

To qualify for Department of Agriculture (DA) programs like the Rice Competitiveness Enhancement Fund (RCEF), you must meet these criteria:

- **RSBSA Registration:** You must be listed in the Registry System for Basic Sectors in Agriculture. This is the **primary requirement** for all cash aid (RFFA), free seeds, and fertilizer vouchers.
- **Land Size:** For direct cash assistance (RFFA), you must till **2 hectares or less** of rice land.

- **Location:** You must be farming in one of the **57 target provinces** identified under RCEF.
- **FCA Membership:** For mechanization and large-scale equipment aid, you must be a member of a **DA-accredited** Farmers' Cooperative or Association (FCA).
- **Proof of Farming:** For those not yet in the RSBSA, a **certificate from the local government** or municipal agriculture office may be required for initial verification.

1. Prepare Documentary Requirements

Gather the following documents before visiting your local office:

- **Accomplished RSBSA Enrollment Form:** Available for free at your municipal/city office.
- **Latest 2x2 ID Picture:** One (1) piece.
- **Valid Identification:** Original and one (1) photocopy of a government-issued ID (e.g., National ID, SSS, GSIS, Voter's ID, or Driver's License).
- **Proof of Farming/Land Status:**
- **For Landowners:** Original and photocopy of Land Title, Certificate of Land Ownership Award (CLOA), or Tax Declaration.
- **For Tenants/Lessees:** Proof of agreement with the landowner or a **Barangay Certification** stating that you are an actual tiller.

2. Visit Your Municipal or City Agriculture Office (C/MAO)

Go to the agriculture office where your farm is located.

1. **Request an Enrollment Form:** If you haven't pre-filled one, ask for a copy.
2. **Submit for Interview & Verification:** An officer will conduct a short interview to verify your farm profile, including crop types (e.g., rice) and farm size.
3. **Biometrics and Signature:** Your photo, signature, and sometimes biometrics will be captured digitally during this step.

3. Obtain Barangay and CAFC/MAFC Signatures

The form often requires validation from local community leaders:

- **Barangay Chairman:** Must sign to certify you are a resident and active farmer in the area.
- **CAFC/MAFC Chairman:** A signature from the City or Municipal Agriculture and Fishery Council (CAFC/MAFC) is also typically required.

4. Receive Your Enrollment Stub

Once submitted, you will receive an **RSBSA Enrollment Stub** containing a temporary control number and a DA official stamp.

- **Keep this stub safe.** It serves as your immediate proof of registration to avail of seeds, fertilizers, and cash aid while your permanent ID or profile is being finalized.

5. Verify Your Status Online

You can check if your registration is already encoded in the national database by using the Official RSBSA Finder. Simply enter your name and birthdate to see your current status.

Planting

Method 1: Transplanting (The Traditional Way)

This is the most common practice, where seedlings are raised in a nursery before being moved to the main field.

1. **Seedbed Preparation:** A small portion of the field (about 1/10th the size of the final planting area) is prepared as a nursery.
2. **Sowing in Nursery:** Pre-germinated seeds are sown in the nursery and grown for **20–40 days** until they become sturdy seedlings.
3. **Pulling Seedlings:** Mature seedlings are manually pulled from the nursery bed and tied into bundles.

4. **Final Field Preparation:** The main field is puddled (plowed and harrowed while flooded) until the soil reaches a mud-like consistency.
5. **Transplanting:** Farmers walk backward, planting **2–3 seedlings** per "hill" in straight lines or random patterns. Lines are often created using a wooden guide to ensure even spacing.

Method 2: Direct Seeding (Sabog-Tanim)

In this method, seeds are sown directly into the main field, eliminating the need for a nursery and the labor-intensive pulling and transplanting steps.

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- **Wet Direct Seeding:** Pre-germinated seeds are "broadcasted" (scattered) by hand or sown using a **drum seeder** over puddled, leveled soil.
- **Dry Direct Seeding:** Used mostly in rainfed areas, where dry seeds are sown into dry soil using a seed drill or by broadcasting, then covered with a harrow.

Key Universal Steps

Regardless of the planting method, these critical preparation steps must occur first:

- **Seed Selection & Treatment:** Farmers select high-quality seeds and often soak them in water for **24 hours**, followed by **24–48 hours** of incubation to trigger germination.
- **Land Preparation:** This involves primary **plowing** (to kill weeds and incorporate old stubble) and **harrowing** (to break up clods), followed by **leveling** the field to ensure even water distribution.
- **Bund Maintenance:** Farmers repair the *pilapil* (small dikes or bunds) to retain water in the field throughout the growing season.

Successful rice farming in the Philippines requires a structured calendar of inputs and labor to maximize the yield of *palay* (unmilled rice). Below is the comprehensive breakdown for a typical one-hectare farm.

1. Production Calendar & Management

Timing is critical to ensure nutrients are available when the plant needs them most.

Activity	Timing (Days After Transplanting/Sowing)	Details
Land Prep	14–21 Days Before Planting	Plowing, harrowing, and leveling the field to manage weeds and water.
Basal Fertilizer	0–2 Days Before/At Planting	Apply half of Nitrogen (N), all Phosphorus (P), and all Potassium (K).
Root Boosters	0–7 Days After Planting	Optional; used to encourage early establishment and resilience.
Herbicide	0–3 Days (Pre) / 10–15 Days (Post)	Apply pre-emergence or early post-emergence to young weeds.
1st Top Dress	14–21 Days After Planting	Apply the first split of Nitrogen (Urea) to promote tillering.
Pesticides	Ongoing (Scout Weekly)	Only apply if pest thresholds (e.g., Stem Borers, Hoppers) are met.
Foliars/Fungicides	30–60 Days After Planting	Apply during the reproductive stage; foliars are most effective before flowering.
2nd Top Dress	50–70 Days (Panicle Initiation)	Apply remaining Nitrogen to support grain development.
Harvesting	110–120 Days (Variety Dependent)	Harvest when 80–85% of grains are straw-colored.

2. Resources Needed (Per Hectare)

- **Machineries:**
- **Land Preparation:** Hand Tractor (₱130k) or 4-Wheel Tractor (₱1M) for faster turnaround.
- **Harvesting:** Rice Combine Harvester (₱2M) or manual threshing with a Rice Thresher (₱130k).

- **Water:** Water Pump (₱100k) for irrigation in rainfed areas.
- **Workforce:**
- **Manual Labor:** Requires roughly **15–20 person-days** for transplanting and **10–12 person-days** for manual harvesting.
- **Mechanized:** A harvester requires only **1–2 operators** to finish a hectare in 3–4 hours.

3. Capital & Profits (2024–2025 Estimates)

Financial performance varies by region, but national averages from the Philippine Statistics Authority (PSA) provide a baseline.

- **Total Capital (Production Cost): ₱59,695** per hectare.
- **Labor:** ~₱29,235 (highest cost component).
- **Fertilizer:** ~₱12,000–₱15,000 (varies by price spikes).
- **Seeds:** ~₱4,300 (Inbred) or ~₱7,000 (Hybrid).
- **Gross Returns: ₱95,906** (based on an average yield of 4.1 metric tons/ha at ₱23/kg).
- **Net Profit: ₱36,211** per hectare per cropping season.

Rice farming in the Philippines relies on a strict schedule of chemical and manual inputs to protect yield. Below are the specific brands and seasonal differences for 2024–2026.

Common Agricultural Input Brands

Popular brands available in the Philippines for rice protection and nutrition include:

Herbicides (Weed Control):

- **Pre-emergence (0–3 days after planting):**
  - Bayer Ronstar 25 EC (Oxadiazon),
  - Syngenta Sofit 300 EC (Pretilachlor), and
  - Saathi Gold

- **Post-emergence (15–30 days after planting):**

  Nomini Gold (Bispyribac-sodium),

  Syngenta Repivox, and

  2,4-D Amine for broadleaf weeds.

Insecticides (Pest Control):

Stemborers & Hoppers:

Corteva Pexalon 106 SC,

Syngenta Alika 247 ZC,

FMC Prevathon SC,

and Virtako.

General Pests:

Cymbush 5 EC (Cypermethrin) and

Karate 2.5 EC (Lambda-cyhalothrin).

Fungicides & Foliars:

- **Disease Control:**

  Bayer Nativo 75 WG (for Blast and Sheath Blight),

  Syngenta Armure, and

  Folicur

Foliar Nutrition:

YaraVita range and

Adaboost Yield Enhancer

Granular Fertilizers:

Standard NPK: YaraMila Unik 16-16-16 and

Nitrogen Source: YaraVera Viking Ship Urea + Zn

Dry Season vs. Wet Season Comparison

The **Dry Season (DS)** is generally more profitable but riskier due to water needs, while the **Wet Season (WS)** faces higher typhoon risks.

Metric	Dry Season (Jan–May)	Wet Season (June–Oct)
Yield Potential	**Higher** (0.5 to 1.5 tons more per ha) due to more solar radiation.	**Lower** due to cloud cover and storm damage risks.
Pest/Disease Pressure	Higher risk of **Brown Planthoppers** and Stemborers.	Higher risk of **Fungal Diseases** (Blast, Sheath Blight) due to humidity.
Production Cost	**Higher** due to increased irrigation/pumping costs.	**Lower** on irrigation but higher on pesticide applications.
Net Profit	Generally **Higher** per kilogram harvested.	Can be **Lower** if typhoons cause significant lodging or grain loss.

Summary of Capital and Profits

- **Average Capital: ₱59,695** per hectare per season.
- **Average Net Profit: ₱36,211** per hectare per season.
- **Top Region for Profit: Central Luzon**, often reaching over **₱58,000** in net returns.
- **Fertilizer Prices:** A 50kg bag of Complete Fertilizer (14-14-14) is currently averaging around **₱1,671** (as of Feb 2026).

Current Local Rice Prices (March 2026)

In BUKIDNON, prices for locally milled rice are competitive compared to national averages, reflecting its status as a major production hub in Northern Mindanao.

- **Milled Rice (Retail in Valencia City):**
  - **Valencia White Rice:** ₱1,380 per 25kg sack (approx. **₱55.20/kg**).

    - **Valencia Red Rice:** ₱1,390 per 25kg sack (approx. **₱55.60/kg**).
    - **Well-Milled Rice (Commercial):** ₱63.50 to ₱64.75 per kg depending on brand.
    - **Regular Milled Rice:** Available for as low as **₱350 per 5kg** (₱70/kg) for premium local brands.
- **Farmgate Palay Price:**
- The Department of Agriculture (DA) has set buying prices for dry *palay* at **₱17.00 to ₱21.00 per kilo** to protect farmers during the harvest peak.
- Some areas report lower farmgate prices near **₱13.00/kg** as more supply enters the market.
- **Government Subsidized Rice:** The **₱20-per-kg** rice program is currently active in all 82 provinces, including Bukidnon, for eligible RSBSA-registered farmers and vulnerable sectors.

Rice Pest Management Calendar (Dry Season)

For farmers in Northern Mindanao (Region 10), the dry season requires vigilant scouting for specific pests.

Growth Stage	Key Pests to Watch For	Management Action
Vegetative (0–40 days)	**Golden Apple Snail**, Armyworms, Green Leafhoppers	Install screens at water inlets; use molluscicides if necessary.
Reproductive (40–80 days)	**Stem Borers**, Rice Tungro Virus (carried by hoppers)	Monitor for "deadhearts"; apply systemic insecticides if thresholds are met.
Ripening (80–120 days)	**Rice Bugs** (*Atangya*), Birds	Clean field bunds to remove rat hideouts; use manual "netting" for birds.

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Rats

Integrated Pest Management (IPM) Tips

- **Weekly Scouting:** Walk through the field in a zigzag pattern once a week to check for "whiteheads" or unusual leaf yellowing.
- **Beneficial Insects:** Avoid over-spraying early in the season to protect natural predators like spiders and ladybugs.
- **Proper Timing:** If using chemicals like Pexalon or Virtako, apply them according to the manufacturer's dosage to prevent pest resistance.

Soil analysis

Soil analysis is essential to maximize your rice yield while minimizing fertilizer waste. In **Valencia City, Bukidnon**, you have access to both government-funded free testing and specialized private services.

Where to Get Soil Analysis

Government Services (Free/Low Cost):

Municipal Agriculture Office (MAO) - Valencia City

Located in the Valencia City Hall Complex, this should be your first stop. They can provide **Soil Test Kits (STK)** or help you submit samples to the **Regional Soils Laboratory** in Cagayan de Oro.

- **DA-Bureau of Soils and Water Management (BSWM):** The government has deployed Mobile Soil Laboratories (MSL)
- across Bukidnon to provide **free, rapid soil and water analysis** directly to farmers,

Private/Commercial Testing:

Allied Material Testing Lab - Malaybalay Branch

While often used for construction, they also perform soil investigation surveys in nearby areas like Cabanglasan.

Terms Concrete and Materials Testing Laboratory

Located in **Mailag, Valencia City**, primarily focused on construction but available for specialized geotechnical soil boring if needed.

Step-by-Step Sampling Process

Proper sampling ensures that your fertilizer recommendation is accurate for your specific field.

1. **Divide the Field:** Group your land into "homogeneous units" based on slope, soil color, or past crop performance.
2. **Collect Sub-samples:** In a V-shape or zigzag pattern, collect **10–15 small soil samples** from a depth of **15–20 cm** (the active root zone for rice).
3. **Mix & Quarter:** Combine all sub-samples in a clean bucket. Mix thoroughly, then take about **1 kg** of this composite mixture.
4. **Dry & Label:** Air-dry the soil in the shade (do not use an oven). Place it in a clean plastic bag and label it with your **name, location, and previous crop**.
5. **Submit:** Bring the sample to the Valencia City MAO for processing.

Cost & Technology Options

Traditional Lab Test: Free to **₱500** depending on the complexity (basic N-P-K vs. complete micronutrient profile).

Minus-One-Element Technique (MOET): A pot-based diagnostic tool available through PhilRice that identifies exactly which nutrient is missing from your soil.

Nutrient Manager for Rice (NMR)**:** A mobile app that provides field-specific fertilizer guidelines based on your soil type and target yield.

General Fertilizer Recommendation

For irrigated lowland rice in Northern Mindanao, a balanced application is critical. Research in Bukidnon indicates that the highest yields (5.0–5.90 tons/ha) are consistently achieved with a balanced rate of **60-60-60 kg/ha.**

Standard Bag-Based Guide (Per Hectare)

A common field-level recommendation for the Valencia-Malaybalay area often includes:

- **Complete Fertilizer (14-14-14):** 4 to 6 bags.
- **Urea (46-0-0):** 2 to 4 bags, depending on the season (more in the dry season).
- **Muriate of Potash (0-0-60):** 0.5 to 1 bag, especially during panicle initiation to improve grain weight.

The "Three-Split" Application Schedule

To maximize efficiency and reduce nutrient loss in heavy clay, follow this timing:

Basal (At Planting/Transplanting):

- **Apply all Phosphorus (P) and about 25-30% of Nitrogen (N) and Potassium (K).**
- **Commonly uses 2-3 bags of 14-14-14.**

Early Topdress (14–21 Days After Planting):

- **Apply another 25% of Nitrogen to promote tillering.**
- **Commonly uses 1-2 bags of Urea.**

Panicle Initiation (50–70 Days After Planting):

- **Apply the remaining 50% of Nitrogen and 30% of Potassium.**
- **Commonly uses 1-2 bags of Urea plus 0.5 bag of 0-0-60.**

Key Soil Management Tips for Bukidnon

- **Liming:** Adtuyon Clay is moderately acidic. Applying **lime** (approx. 2–4 tons/ha) is highly recommended by BSWM to improve nutrient availability.

- **Zinc Supplementation:** Many Bukidnon soils are zinc-deficient. Applying **Zinc Sulfate** (5–10 kg/ha) basally can prevent stunted growth.
- **Organic Incorporation:** Since Adtuyon Clay can become very hard when dry, incorporating **rice straw** or organic matter improves soil tilth and water-holding capacity.

Where to Buy in Valencia City

You can use **DA fertilizer vouchers** at several accredited merchants in Valencia:
Valencia Farmers Cooperative (near the City Public Market).

Green Agri-Supply (Lumbo, Valencia City).

Yara Fertilizer Dealers (check local agri-supply stores for the "Viking Ship" brand).

How and when to apply fertilizers, root boosters, herbicide, pesticides, foliars, fungicides, etc.

To maximize your yield on Rice, timing is everything. Here is the specific "when and how" for every input on a typical 115-day rice variety.

The Input Calendar (Days After Transplanting - DAT)

Timing	Input Type	Specific Product Example	How to Apply
0–2 Days	**Herbicide (Pre-E)**	*Ronstar* or *Sofit*	Spray on saturated soil with 2-3cm water.
0–2 Days	**Basal Fertilizer**	*14-14-14 (Complete)*	Broadcast 4 bags/ha before final leveling.
5–7 Days	**Root Booster**	*Humi-K* or *Vigor*	Mix with water; drench or spray base of plants.

10–15 Days	**Herbicide (Post-E)**	*Nomini Gold*	Spray if weeds escaped the first treatment.
18–21 Days	**1st Topdress**	*Urea (46-0-0)*	Broadcast 2 bags/ha to trigger tillering.
35–45 Days	**Pesticide**	*Prevathon* or *Virtako*	Spray if you see "deadhearts" (Stemborer).
50–55 Days	**2nd Topdress**	*Urea + 0-0-60*	Broadcast 2 bags Urea + 1 bag Potash.
55–65 Days	**Foliar & Fungicide**	*YaraVita + Nativo*	Spray at "Boots" stage (before flowering).
85–90 Days	**Late Foliar**	*Adaboost*	Spray to improve grain filling/weight.

Application Masterclass

1. Fertilizers (Granular)

- **How:** Always broadcast (scatter) when the morning dew has dried to prevent leaf burn.
- **Pro Tip:** Keep water levels at **3–5cm** during application to ensure the nutrients dissolve and reach the roots. Do not drain the field for 4 days after applying.

2. Herbicides & Pesticides

- **Herbicide:** Must be applied on **leveled soil**. If the field is bumpy, weeds will grow on the "islands" where the chemical didn't reach.
- **Pesticide:** Never spray on a schedule. **Scout first.** If you see more than 2 stemborer moths per square meter, then you spray.

3. Foliars & Fungicides

- **When:** Apply early morning (6 AM – 9 AM) or late afternoon (4 PM – 6 PM) when the **stomata** (pores) of the leaves are open.
- **Fungicides:** Critical during the "Booting" stage (when the stalk bulges) to prevent **Rice Blast** and **Sheath Blight**, which are common in the humid Bukidnon climate.

4. Root Boosters

- **Why:** In the heavy clay of Valencia, roots can struggle to penetrate deep early on. Humic acid-based boosters help soften the soil around the root zone.

Quick Checklist for Farmers

- **Water Management:** Use **Alternate Wetting and Drying (AWD)**. Let the water naturally subside until you see cracks in the soil, then re-flood. This makes roots grow deeper.
- **Calibration:** Ensure your knapsack sprayer is calibrated. One 16-liter tank should cover approximately 250 square meters.
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1. Harvesting: Timing and Preparation

Harvesting should occur when **80–85% of the grains are straw-colored** (golden yellow).

- **Optimal Timing:**
- **Dry Season:** 28–35 days after heading (flowering).
- **Wet Season:** 32–38 days after heading.
- **Field Preparation:** Drain the field **7–10 days before harvest** to allow the soil to harden, facilitating easier movement for both manual labor and heavy machinery like combine harvesters.
- **Moisture Content (MC):** Aim for **20–25% MC** for mechanical harvesting to prevent grain shattering.

2. Threshing: Methods and Best Practices

Threshing (separating grain from the stalk) must be done **within 24 hours of cutting** to prevent spoilage and grain discoloration caused by heat buildup in the piles.

- **Mechanical (Recommended):** Use a **Throw-in or Axial-flow Thresher**. Ensure the drum speed is correctly set according to the manual to avoid grain cracking.
- **Manual:** Common in smaller plots, involving "beating" stalks against a wooden frame (*hampas*) or trampling (*gi-ok*).
- **Cleaning:** Use the thresher's built-in blower or a separate winnower to remove "chaff" (empty grains) and straw.

3. Post-Harvest Preparations & Storage

- **Drying:** Immediately sun-dry on tarpaulins or use mechanical dryers to reach a **14% MC** for milling or **12% MC** for seed storage.
- **Stacking:** Store bags on **wooden pallets** at least 1 meter away from walls to allow air circulation and prevent moisture from the floor from rotting the grains.

4. Local Facilities in Valencia City

Valencia City is benefiting from significant agricultural modernization:

- **Grains Center:** A new **₱101.5-M Grains Center** in **Brgy. Sinabuagan** features high-capacity rice mills and mechanical recirculating dryers to help farmers process their harvest efficiently.
- **National Food Authority (NFA):** The NFA GID Valencia on Sayre Highway is a primary site for selling and storing *palay* at government-guaranteed prices.
- **Milling:** Commercial traders like YP Rice Trading in Poblacion provide local milling and buying services.